Again on off-policy methods with approximations

Master Degree in Control Systems Engineering 2024-2025
University of Padova

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Just a clarification…motivated by a question of one of your colleague

Episodic Semi-gradient Sarsa for estimating $\hat{q} \approx q_*$

Input : $\hat{q} : \mathcal{S} \times \mathcal{A} \times \mathbb{R}^d \to \mathbb{R}$ $(\hat{v}(\text{terminal},\cdot) = 0)$, step-size α

Algorithm parameters : step-size α , small $\varepsilon > 0$

Initialize value-function weights $\mathbf{w} \in \mathbb{R}^d$ arbitrarily (e.g. $\mathbf{w} = 0$)

Repeat (for each episode) :

$S, A \leftarrow$ initial state and action of episode (e.g., ε-greedy)

Repeat (for each step of the episode) :

Take action A, observe R, S'

If S' is terminal

$$\mathbf{w} \leftarrow \mathbf{w} + \alpha \left[R - \hat{q}(S, A, \mathbf{w})\right] \nabla \hat{q}(S, A, \mathbf{w})$$

Go to next episode

Choose A' as a function of $\hat{q}(S', \cdot, \mathbf{w})$ (e.g., ε- greedy)

$$\mathbf{w} \leftarrow \mathbf{w} + \alpha \left[R + \gamma \hat{q}(S', A', \mathbf{w}) - \hat{q}(S, A, \mathbf{w})\right] \nabla \hat{q}(S, A, \mathbf{w})$$

$S \leftarrow S'$

$A \leftarrow A'$

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Remember…convergence of prediction methods

On/Off-Policy	Algorithm	Table Lookup	Linear	Non-Linear
On-Policy	MC	✓	✓	✓
	TD(0)	✓	✓	✗
	TD(λ)	✓	✓	✗
Off-Policy	MC	✓	✓	✓
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Convergence of control methods

Algorithm	Table Lookup	Linear	Non-Linear
Monte-Carlo Control	✓	(✓)	✗
Sarsa	✓	(✓)	✗
Q-learning	✓	✗	✗

(✓) = chatters around near-optimal value function

Today’s lecture

- Why off-policy methods do not work (counterexamples + deadly triad)
- Linear value-function geometry – TD gradient (only mentioned)

Off-policy methods

π : **target policy** b : **exploring policy**

in off-policy learning we seek to learn a value function for a target policy π, given data due to a different behavior policy b

Many of the tabular off-policy algorithms use the *importance sampling ratio :*

$$\rho_t \doteq \rho_{t:t} = \frac{\pi(A_t|S_t)}{b(A_t|S_t)}$$

$$\rho_{t:T-1} \doteq \frac{\prod_{k=t}^{T-1} \pi(A_k|S_k)p(S_{k+1}|S_k, A_k)}{\prod_{k=t}^{T-1} b(A_k|S_k)p(S_{k+1}|S_k, A_k)} = \prod_{k=t}^{T-1} \frac{\pi(A_k|S_k)}{b(A_k|S_k)}$$

$$\rho_{t:h} \doteq \prod_{k=t}^{\min(h,T-1)} \frac{\pi(A_k|S_k)}{b(A_k|S_k)}$$

Off-policy methods with function Approx. : semi-gradient alg.

Natural extension : *To convert them to semi-gradient form, we simply replace the update to an array (V or Q) to an update to a weight vector ($\mathbf{w}$), using the approximate value function ($\hat{v}$ or $\hat{q}$) and its gradient.*

Examples : the one-step, state-value, semi-gradient off-policy TD(0)

$$\mathbf{w}_{t+1} \doteq \mathbf{w}_t + \alpha \rho_t \delta_t \nabla \hat{v}(S_t, \mathbf{w}_t)$$

where

$$\delta_t \doteq R_{t+1} + \gamma \hat{v}(S_{t+1}, \mathbf{w}_t) - \hat{v}(S_t, \mathbf{w}_t)$$

and

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Examples : the one-step, action values, semi-gradient Expected Sarsa

$$\mathbf{w}_{t+1} \doteq \mathbf{w}_t + \alpha \delta_t \nabla \hat{q}(S_t, A_t, \mathbf{w}_t)$$

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$$\delta_t \doteq R_{t+1} + \gamma \sum_a \pi(a|S_{t+1}) \hat{q}(S_{t+1}, a, \mathbf{w}_t) - \hat{q}(S_t, A_t, \mathbf{w}_t)$$

Notice: no use of importance sampling

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These algorithms do not converge as robustly as they do under on-policy training

The extension to function approximation turns out to be significantly different and harder for off-policy learning than it is for on-policy learning

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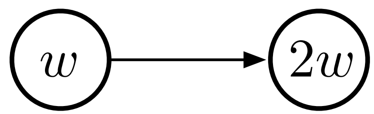
Remember : The on-policy distribution is important to the stability of semi-gradient methods.

Examples of Off-policy Divergence

To establish intuitions, we start by considering a very simple example.

Suppose, as part of a larger MDP, there are two states whose estimated values are of the functional form w and 2w, where the parameter vector w consists of only a single component w.



expressions inside the two circles indicate the two state's values

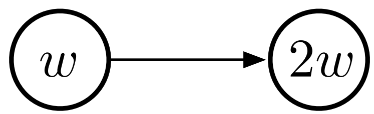
This occurs under linear function approximation if the feature vectors for the two states are each simple numbers (single-component vectors), in this case 1 and 2.

Only one action is possible in this state : a deterministic transition to state 2 with reward 0

Examples of Off-policy Divergence

Suppose initially $w = 10$

The transition will then be from a state of estimated value 10 to a state of estimated value 20.



$$\delta_t = R_{t+1} + \gamma \hat{v}(S_{t+1}, \mathbf{w}_t) - \hat{v}(S_t, \mathbf{w}_t)$$

$$\mathbf{w}_{t+1} \doteq \mathbf{w}_t + \alpha \rho_t \delta_t \nabla \hat{v}(S_t, \mathbf{w}_t)$$

If γ is nearly 1, then the TD error will be nearly 10, and, if $\alpha = 0.1$, then w will be increased to nearly 11 in trying to reduce the TD error.

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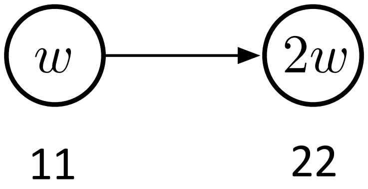

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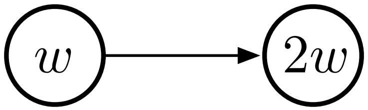

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If the transition occurs again, then it will be from a state of estimated value ≈11 to a state of estimated value ≈22, for a TD error of ≈11 **- larger, not smaller, than before**.

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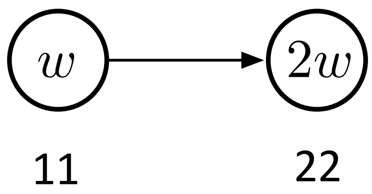

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If the transition occurs again, then it will be from a state of estimated value ≈11 to a state of estimated value ≈22, for a TD error of ≈11 **- larger, not smaller, than before**.

It will look even more like the first state is undervalued, and its value will be increased again, this time to ≈12.1. This looks bad, and in fact with further updates w will diverge to infinity.

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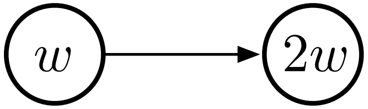
Let us see more carefully what seen in the previous slide at the sequence of updates



The TD error on a transition between the two states is

$$\delta_t = R_{t+1} + \gamma \hat{v}(S_{t+1}, \mathbf{w}_t) - \hat{v}(S_t, \mathbf{w}_t) = 0 + \gamma 2 w_t - w_t = (2\gamma - 1) w_t$$

and the off-policy semi-gradient TD(0) update is

$$\begin{aligned} w_{t+1} = w_t + \alpha \rho_t \delta_t \nabla \hat{v}(S_t, w_t) &= w_t + \alpha \cdot 1 \cdot (2\gamma - 1) w_t \cdot 1 \\ &= \big(1 + \alpha(2\gamma - 1)\big) w_t \end{aligned}$$

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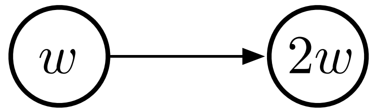

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Note that the importance sampling ratio, ρ_t, is 1 on this transition because there is only one action available from the first state, so its probabilities of being taken under the target and behavior policies must both be 1.

Examples of Off-policy Divergence

$$w_{t+1} = \big(1 + \alpha(2\gamma - 1)\big)w_t$$

If $1 + \alpha(2\gamma - 1) > 1$ then the system is unstable and w will go to positive or negative infinite depending on its initial value

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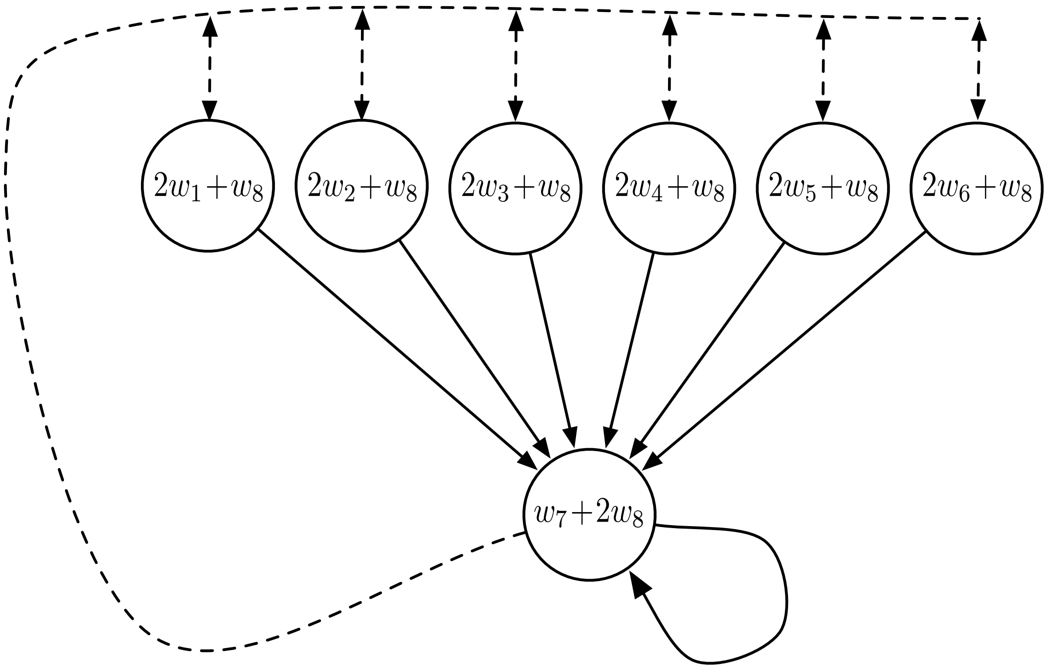
Key to this example is that the one transition occurs repeatedly without w being updated on other transitions. *This is possible under off-policy training because the behavior policy might select actions on those other transitions which the target policy never would.* For these transitions, ρ_t would be zero and no update would be made.

Examples of Off-policy Divergence

This simple example communicates much of the reason why off-policy training can lead to divergence, but it is not completely convincing because it is not complete—it is just a fragment of a complete MDP.

Can there really be a complete system with instability? A simple complete example of divergence is *Baird's counterexample*.

Examples of Off-policy Divergence



Seven state, two-action MDP

The dashed action takes the system to one of the six upper states with equal probability, whereas the solid action takes the system to the seventh state

$\pi(\text{solid}|\cdot) = 1$

$b(\text{dashed}|\cdot) = 6/7$

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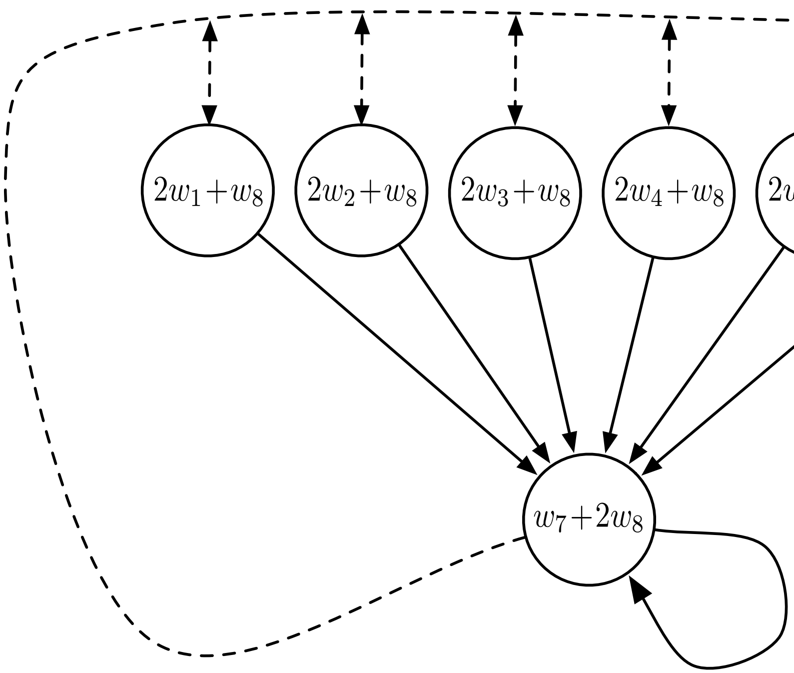

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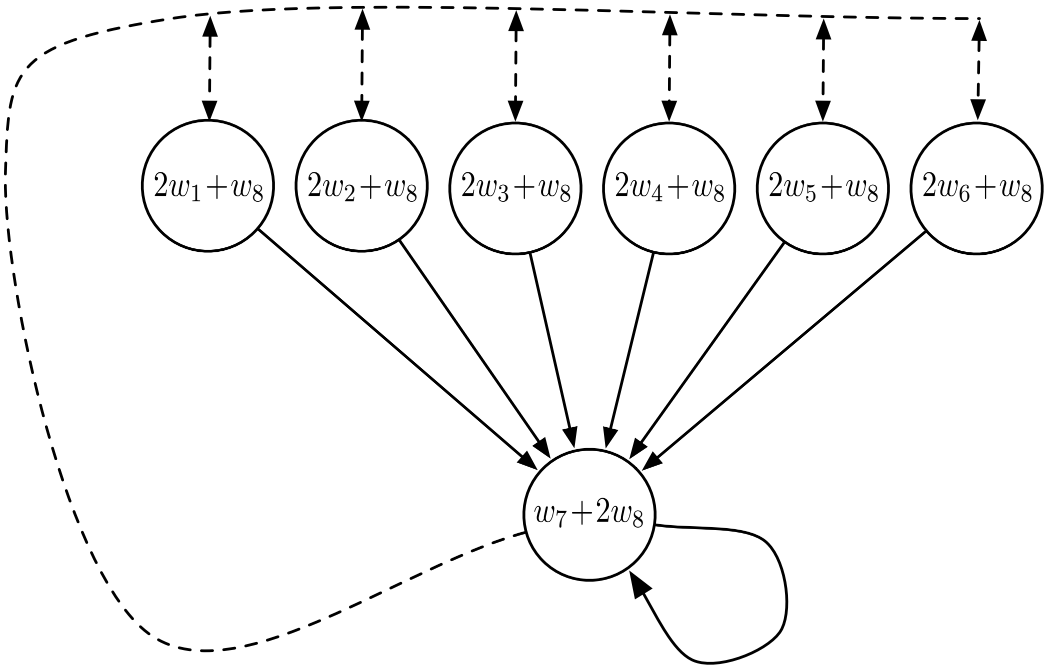

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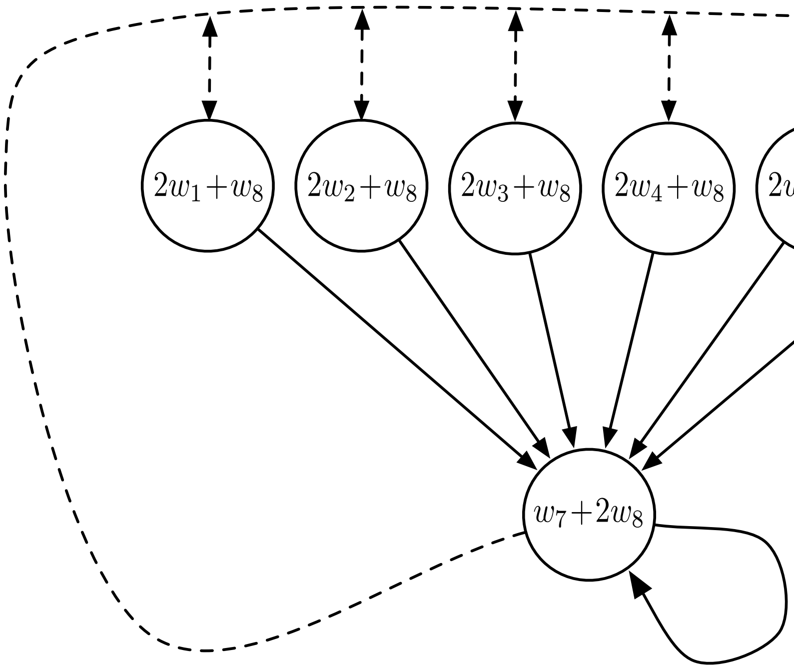
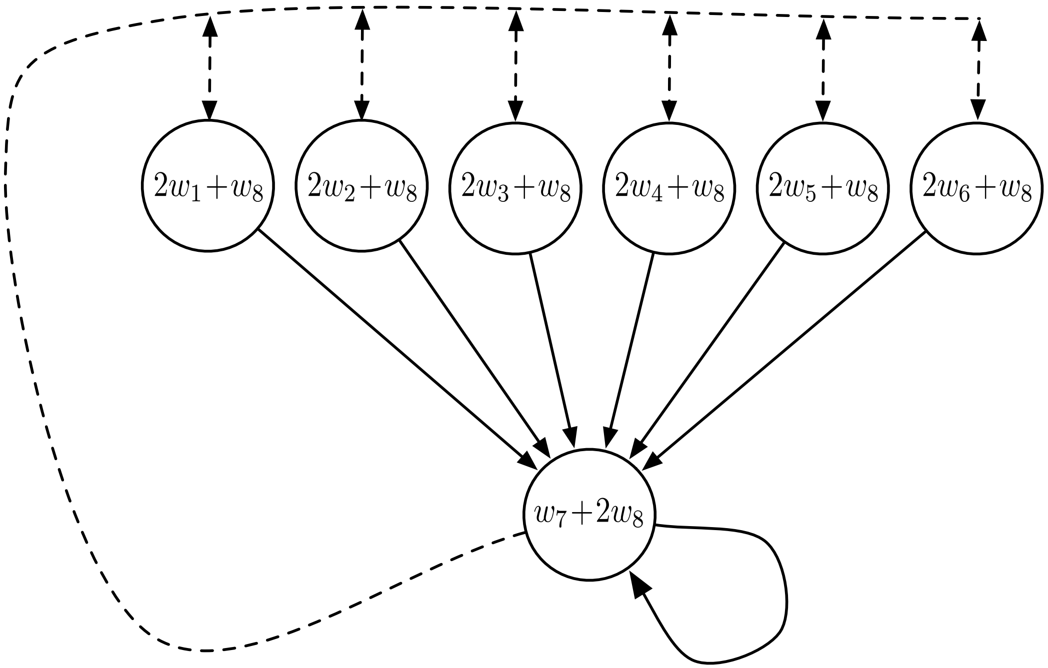


Consider estimating the state-value under the linear parameterization indicated by the expression shown in each state circle.

Examples of Off-policy Divergence



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For example, the estimated value of the leftmost state is $2w_1 + w_8$, where the subscript corresponds to the component of the overall weight vector $w \in \mathbb{R}^8$; this corresponds to a feature vector for the first state being $x(1) = (2,0,0,0,0,0,0,1)^T$

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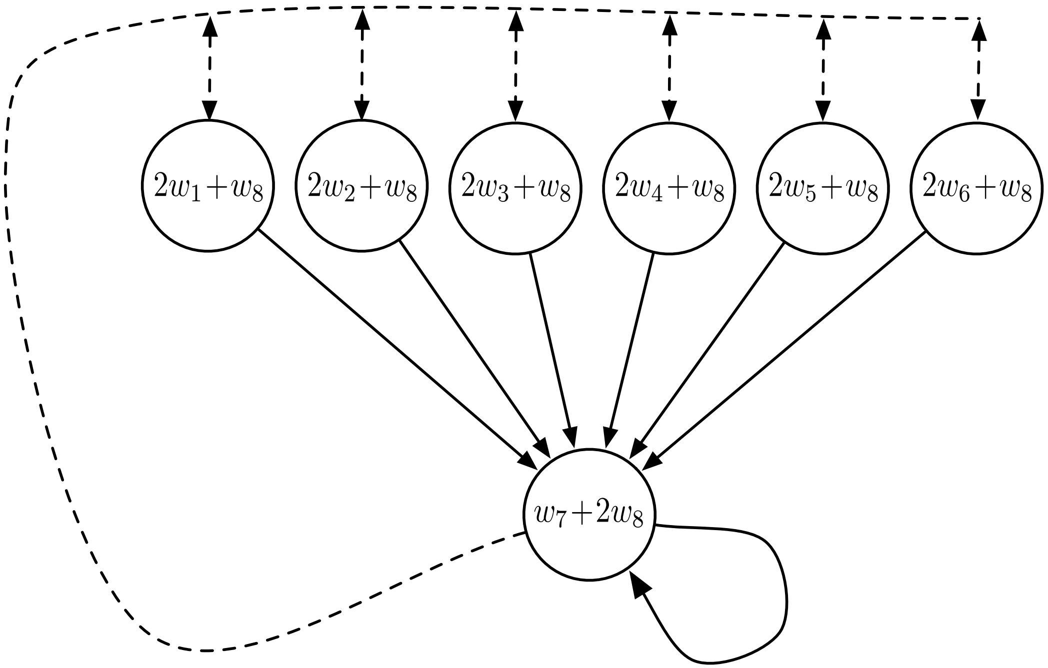

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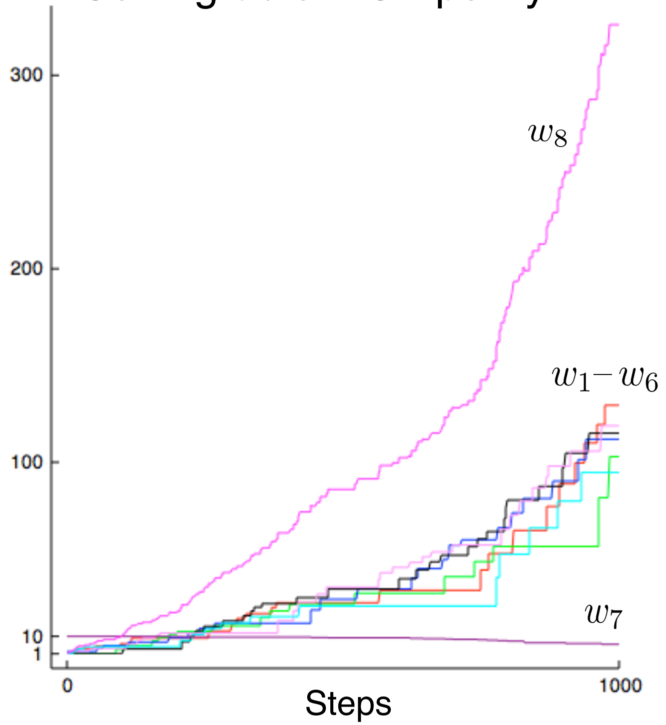
The reward is zero on all transitions, so the true value function is $v_\pi(s) = 0$, for all s, which can be exactly approximated if $\mathbf{w} = 0$.

Examples of Off-policy Divergence

If we apply semi-gradient TD(0) $\mathbf{w}_{t+1} \doteq \mathbf{w}_t + \alpha\rho_t\delta_t\nabla\hat{v}(S_t,\mathbf{w}_t)$ to this problem then the weights diverge to infinity

Semi-gradient Off-policy TD



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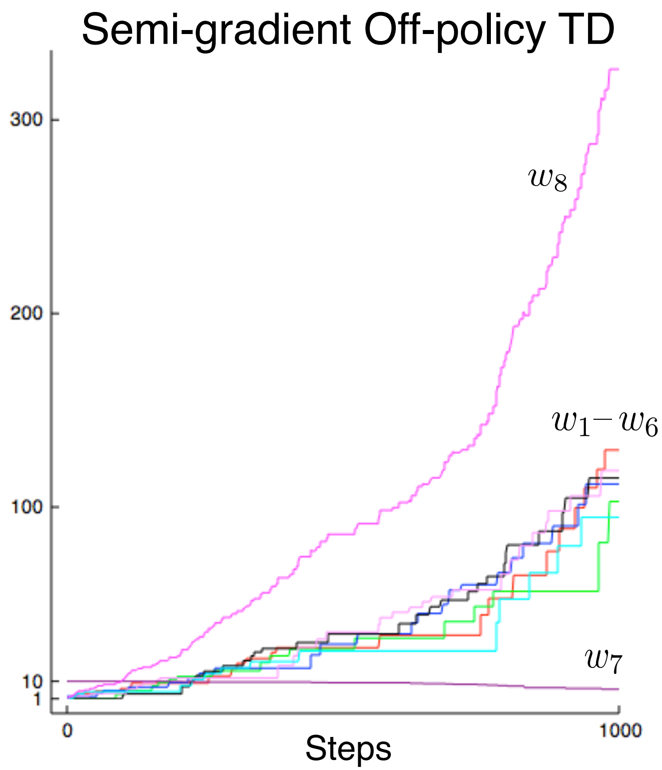

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The example shows that even the *simplest combination of bootstrapping and function approximation can be unstable* if the updates are not done according to the on-policy distribution.

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This is cause for concern because otherwise Q-learning has the best convergence guarantees of all control methods. Considerable effort has gone into trying to find a remedy to this problem or to obtain some weaker, but still workable, guarantee.

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For example, it may be possible to guarantee convergence of Q-learning as long as the *behavior policy is sufficiently close to the target policy*, for example, when it is the ε-greedy policy.

To the best of our knowledge, Q-learning has never been found to diverge in this case, but there has been no theoretical analysis.

The deadly triad

The danger of instability and divergence arises whenever we combine all of the following three elements

- **Function approximation**
- **Bootstrapping**
- **Off-policy training**

The deadly triad

If any two elements of the deadly triad are present, but not all three, then instability can be avoided.

It is natural, then, to go through the three and see if there is any one that can be given up ???

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- Of the three, *function approximation most clearly cannot be given up*. We need methods that scale to large problems and to great expressive power

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Losses in data efficiency

- Bootstrapping guarantees *faster learning*

The deadly triad

- Off-policy learning; can we give that up? *On-policy methods are often adequate*.

For model-free reinforcement learning, one can simply use Sarsa rather than Q-learning. *Using off-policy learning is considered as an appealing convenience but not a necessity*.

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For model-free reinforcement learning, one can simply use Sarsa rather than Q-learning. *Using off-policy learning is considered as an appealing convenience but not a necessity*.

However, off-policy learning is essential to other anticipated use cases, cases that we do not discuss in this class but may be important to the larger goal of creating a powerful intelligent agent. In these use cases, *the agent learns not just a single value function and single policy, but large numbers of them in parallel.*

Gradient Temporal-Difference learning

- TD does not follow the gradient of *any* objective function
- This is why TD can diverge when off-policy or using non-linear function approximation
- Gradient TD follows true gradient of projected Bellman error

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Convergence of control algorithms

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(✓) = chatters around near-optimal value function

Material

The material illustrated in this second part of the lecture can be found in the first three paragraphs of Chapter 11.

In the book there are more descriptions and thoughs. For the final examination pay attention to what seen in the slides used for the lecture (as usual concentrate on the algorithms!)